

Chapter 11 - Functions

4 pts

1. Let $A = \{1, 2, 3, 4, 5, 6\}$. In each case, give an example of a function $f : A \rightarrow A$ with the indicated properties, or explain why no such function exists.

(a) f is bijective, but is not the identity function $f(x) = x$.

(b) f is neither one-to-one nor onto.

(c) f is one-to-one, but not onto.

(d) f is onto, but not one-to-one.

4 pts

2. In each case, give an example of a function $f : \mathbb{N} \rightarrow \mathbb{N}$ with the indicated properties, or explain why no such function exists.

(a) f is bijective, but is not the identity function $f(x) = x$.

(b) f is neither one-to-one nor onto.

(c) f is one-to-one, but not onto.

(d) f is onto, but not one-to-one.

4 pts

3. In each case, give an example of a function $f : \mathbb{Z} \rightarrow \mathbb{Z}$ with the indicated properties, or explain why no such function exists.

(a) f is bijective, but is not the identity function $f(x) = x$.

(b) f is neither one-to-one nor onto.

(c) f is one-to-one, but not onto.

(d) f is onto, but not one-to-one.

Chapter 12 - Special Functions

3 pts

1. Let $f(x) = x^3$ and $g(x) = 3^x$. Notice that $f(1) < g(1)$ and $f(3) = g(3)$. Does g ever catch up with f again, or does f always stay ahead of g ?

3 pts

2. Write $\frac{5}{2} \ln 5 - 4 \ln 3$ as a single logarithm.

3 pts

3. Write $\lg 1 + \lg 2 + \lg 3 + \lg 4 + \lg 5 + \lg 6$ as a single logarithm.

3 pts

4. Write $\sum_{k=1}^n \lg n$ as a single logarithm.